

WJ Boshoff

Software Engineer

Email: wjbosdev@gmail.com Phone: 079 873 7177 Portfolio: wj-bos.github.io

Location: 79 Rae Frankel Street, Brackenhurst, 1448 Alberton

PROFESSIONAL SUMMARY

Full-stack Software Engineer with proven expertise spanning enterprise fintech systems and production mobile applications, delivering solutions across the full software development lifecycle. Specialized in building scalable microservices architectures, cloud infrastructure automation, complex financial integrations, AI-powered features that augment core business workflows, real-time media streaming, and cross-platform mobile apps in Kotlin and Flutter. Demonstrated ability to architect and implement mission-critical systems including authentication platforms, observability infrastructure, and business intelligence solutions. Strong track record of taking ownership from system design through production deployment, with deep technical proficiency across the JVM ecosystem, .NET, mobile (Kotlin / Flutter), AWS cloud services, CI/CD automation, and distributed systems.

PROFESSIONAL EXPERIENCE

Software Engineer, IdealSolution | 01/2025 – Present

Centurion, Gauteng

Mobile Development — World Rugby Match Tracking Application

- Helped build a production Kotlin Android application used by World Rugby to track live game progression across matches and tournaments, with real-time synchronization of the game timeline across connected devices
- Engineered low-latency live audio streaming and time-synchronization using WebRTC and HLS, keeping in-app match audio aligned with on-field game state under live broadcast and officiating conditions
- Implemented real-time match timers, player time tracking, card management (yellow / red), and per-tournament event logging — built for reliable use during live matches
- Led a full rebuild of the companion Admin portal in Angular, redesigning the architecture and UI to manage matches, tournaments, and live event data feeding the mobile application
- Worked across the native Android stack in Kotlin, applying Material Design patterns and Android lifecycle best practices

Financial Systems & Third-Party Integrations

- Architected and deployed production integrations with multiple financial service providers including Sage (contract synchronization) and RealPay (payment processing, client onboarding, mandate management, debit order automation)
- Built robust data transformation pipelines handling JSON-to-XML conversions for legacy system compatibility, ensuring seamless cross-platform financial operations
- Engineered automated financial workflows that eliminated manual processing bottlenecks and reduced error rates in client account management

Business Intelligence & Data Analytics

- Designed and implemented financial reporting dashboards from scratch using AWS QuickSight, delivering real-time insights into business operations and financial metrics
- Created data integrity microservice using Spring Boot with dynamic Spring IoC container configuration, enabling automated validation across distributed services
- Deployed intelligent monitoring system with role-based email alerting across three AWS environments (ECS + Fargate), significantly improving data quality and operational visibility

AI Integration & Intelligent Automation

- Integrated large language model APIs into enterprise workflows, designing prompt engineering strategies and structured-output contracts to deliver reliable AI-assisted features within production Spring Boot services
- Built AI-powered automation components for document parsing, data extraction, and natural-language querying of business data, reducing manual operational workload
- Established patterns for safe LLM integration including input validation, output schema enforcement, retry/fallback strategies, and cost/latency monitoring

Cloud Infrastructure & DevOps

- Owned end-to-end AWS ECS deployments including ECR image management, task definition configuration, service orchestration, and custom VPC networking design
- Implemented comprehensive CI/CD pipelines using Jenkins and Azure DevOps, automating build, test, and deployment workflows
- Configured blue-green deployment strategies using AWS CodeDeploy, enabling zero-downtime releases and rapid rollback capabilities

Observability & Distributed Systems

- Architected full-stack distributed tracing solution using OpenTelemetry, AWS Distro for OpenTelemetry (ADOT), and SigNoz on ECS
- Implemented end-to-end log aggregation and correlation across microservices architecture, dramatically reducing mean time to resolution (MTTR) for production incidents
- Established observability best practices including structured logging, metrics collection, and distributed tracing standards

Authentication & Security Architecture

- Led critical migration initiative from AWS Cognito to custom-built authentication system, architecting user management, role-based access control (RBAC), and multi-tenant infrastructure
- Designed and implemented JWT-based authentication with token refresh mechanisms and role hierarchy management
- Engineered row-level security (RLS) at the database layer, ensuring data isolation across tenants and implementing fine-grained access controls
- Built multi-tenant branch management system supporting complex organizational hierarchies and permission inheritance

Enterprise Application Development

- Contributed to full-stack development of comprehensive car rental management platform encompassing lead generation, contract lifecycle, fleet tracking, and financial reporting
- Designed robust CSV import/export functionality with multi-layer validation, error handling, and data quality checks, processing high-volume datasets reliably
- Took ownership of complete feature delivery from requirements analysis through QA and production deployment
- Collaborated in agile environment, conducting code reviews and maintaining high code quality standards

EDUCATION

Certificate of Higher Education: Software Development, CTU Training Solutions | 01/2023 – 11/2024

Graduated May 2025. Comprehensive curriculum covering software architecture, database design, full-stack development, and modern development practices. Practical experience with enterprise frameworks, cloud platforms, and CI/CD automation. *Potchefstroom*

Certificate of Higher Education: Networking, CTU Training Solutions | 01/2022 – 11/2022

Foundation in network architecture, security protocols, infrastructure design, and systems administration. *Potchefstroom*

TECHNICAL EXPERTISE

Languages & Frameworks

Java, Spring Boot (Core, Security, Data JPA), C#, .NET / ASP.NET Core, Kotlin, Dart / Flutter, TypeScript, Angular, React, Go

Real-Time Media & Streaming

WebRTC, HLS, Live Audio Streaming, Time Synchronization, Low-Latency Media Delivery

DevOps & CI/CD

Jenkins, GitHub Actions, Docker, Infrastructure as Code (Terraform), Blue-Green Deployments, Automated Testing Pipelines

Security & Authentication

JWT Authentication, OAuth 2.0, Role-Based Access Control (RBAC), Multi-Tenant Architecture, Data Encryption

Databases & Data

PostgreSQL, MySQL, Oracle, SQL Optimization, Row-Level Security, Database Design, Data Migration

Software Engineering

Microservices Architecture, System Design, Design Patterns, Agile/Scrum, Git Version Control, Code Review

Mobile Development

Kotlin (Android), Flutter, Dart, Material Design, Push Notifications (FCM), Native Platform Channels, App Store / Play Store Deployment, Offline-First Architecture

AI & Intelligent Systems

LLM API Integration, Prompt Engineering, Structured Output Contracts, Retrieval-Augmented Generation (RAG) Patterns, AI-Assisted Workflow Automation

Observability & Monitoring

OpenTelemetry, AWS Distro for OpenTelemetry (ADOT), SigNoz, Distributed Tracing, Structured Logging, CloudWatch

Cloud & Infrastructure

AWS (ECS, Fargate, ECR, CodeDeploy, VPC, IAM, QuickSight, CloudWatch), Azure DevOps, Linux Server Administration, Docker, Container Orchestration

Integration & Messaging

REST APIs, ActiveMQ, Message Queues, JSON/XML Processing, Third-Party API Integration

LANGUAGES

Afrikaans – Native English – Fluent

PROJECTS & INITIATIVES

Boshetto — Music Theory Learning Platform

boshetto.com · Web · TanStack Start, TypeScript, PostgreSQL, Drizzle, tonal.js, Tone.js, AlphaTab

Music theory learning platform built around the principle of "learn music theory the way it actually works," targeted at guitarists. Features include an interactive SVG fretboard, structured curriculum with prerequisites, ear-training exercises, spaced-repetition

scheduling, runtime audio synthesis via Tone.js, and tab rendering with AlphaTab. Backed by PostgreSQL via Drizzle for progress persistence.

Yap — Voice-First Productivity App

Mobile · Flutter, .NET 10 / ASP.NET Core Minimal API, PostgreSQL / pgvector, Whisper, Gemini 2.5 Flash

Cross-platform mobile productivity app that converts spoken voice notes into structured actions — tasks, calendar events, reminders, emails, shopping lists, and expense entries. Architected the full stack: Flutter frontend, .NET 10 / ASP.NET Core Minimal API backend, Whisper for transcription, and Gemini 2.5 Flash for both structured extraction and generative outputs, with pgvector for semantic retrieval over the user's entry history.

Koinon — AI-Powered Bible Study App

Mobile · Flutter, Rust, PostgreSQL, LLM Ingestion & Generation

Cross-platform mobile Bible study app built in Flutter with a Rust-backed service layer and PostgreSQL. Designed and implemented an AI ingestion-and-generation pipeline that uses the books of the Bible as primary context to produce structured, curated study plans tailored to the user. Includes Firebase Cloud Messaging push notifications, generated cover art pipeline, and a typography / branding system designed for long-form reading.

Local Cloud Control Plane

github.com/WJ-Bos/local-cloud · Java 21, Spring Boot, React, Terraform, Docker, PostgreSQL

Self-built platform that mimics AWS RDS-style database provisioning on local hardware. Users provision and manage PostgreSQL, MySQL, MariaDB, MongoDB, and Redis instances through a React UI; the Spring Boot backend generates Terraform configurations per instance and delegates container lifecycle to the Docker provider. Implements full lifecycle management (create, start, stop, update, destroy), live container log streaming, version selection per engine, automatic port allocation, and encrypted credential storage in a platform metadata database.

JWT Authentication Service

Developed production-ready authentication microservice with token-based access control, refresh token rotation, and secure session management.

Self-Hosted VPS Infrastructure

Architected and deployed personal cloud platform on Linux VPS using Dokploy orchestration. Implemented Dockerized application deployments with automated CI/CD workflows, gaining advanced experience in server administration, container networking, and infrastructure automation.